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China Humanoid Robots

US FCC to Ban Foreign-Made Humanoid and Quadruped Robots in the US

CITI'S TAKE

On 28 July, the US Federal Communications Commission (FCC) added advanced robotic devices, including humanoids and quadrupeds, to its [Covered List](#), meaning that foreign-made advanced robotic devices will be banned from being imported into the US due to national security concerns ([link](#)). Given China's well-established robotic industry, and US-China trade and political disputes, we view the development as particularly relevant for Chinese companies. Chinese robotic makers with high exposure to US customers are likely to bear the brunt. However, we believe this could potentially benefit Chinese component makers that are already in the supply chain of US humanoid robot companies, such as [Hengli Hydraulic \(601100.SS\)](#) and [Leader Drive \(688017.SS\)](#), since this ban suggests that US robotic companies should gain more market share in the US market. In this note, we list key Chinese component makers that supply to leading humanoid robot companies.

Our view on geopolitical risks – We believe Chinese humanoid robot makers would be impacted more by the geopolitical risk (e.g., restrictions or sanctions) than robot component markets, as humanoid robots are often equipped with sensors that collect and maintain sensitive data on their surroundings. Chinese humanoid robot component makers, on the other hand, should face lower or limited geopolitical risk as many of them, especially those in the Tesla Optimus supply chain, have moved their humanoid robot production lines to ASEAN countries (Thailand, in particular), Mexico, or even to the US to avoid directly exporting products from China to the US. For example, Hengli Hydraulic (601100.SS) is gradually moving its planetary roller screw production lines to Mexico; [Leader Drive \(688017.SS\)](#) has established a humanoid actuator JV with Minth (425.HK) in the US; and [Rongtai \(603119.SS\)](#) and [Veichi \(688698.SS\)](#) have formed a humanoid robot micro ball screw and motor JV in Thailand for their key US customer.

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Figure 1. Key humanoid robot component suppliers that we are aware of

Key Component	Tesla (TSLA.O)' Potential Suppliers	UBTECH's (9880.HK) Suppliers	AgilBot/Zhiyuan's Suppliers	Unitree's Suppliers
GPU	Nvidia (NVDA.O) Tesla - SOC (TSLA.O)	Nvidia (NVDA.O)	Nvidia (NVDA.O)	Nvidia (NVDA.O)
CPU	Arm (ARM.O) Leader Drive (688017.SS) Sling (301550.SZ)	Intel (INTC.O)	Intel (INTC.O)	Intel (INTC.O)
Harmonic Reducer	Kedali (002850.SZ) HDS (6324.T) Nidec (6594.T)	Leader Drive (688017.SS)	Leader Drive (688017.SS)	
Planetary Reducer		NEWSTART (unlisted)	ZhongDa Leader (002896.SZ)	ZhongDa Leader (002896.SZ)
Modified RV Reducer for Waist	Shuanghuan (002472.SZ)			
Planetary Roller Screw	Schaeffler (SHA0n.DE) Hengli Hydraulic (601100.SS) Seenpin (unlisted)			
Frameless Torque Motor for Arm	Kollmorgen (unlisted) Moons' Electric (603728.SZ)	Kinco (688160.SS)	Fulling Motor (unlisted) Kinco (688160.SS) Wolong (600580.SS)	
Brushless Motor for Wrist	Zhaowei (2692.HK / 003021.SZ)			
3D Vision System	Tesla (TSLA.O) Orbbec (688322.SS)	Orbbec (688322.SS)	Orbbec (688322.SS)	RealSense (unlisted) Siyi Technology (unlisted)
Battery		BYD / FinDreams (1211.HK)		
6-axis Force Sensor		Kunwei Tech (unlisted) Sri Sensor (unlisted)		
Dexterous Hand - Coreless Motor	Veichi (688698.SS)	Zhaowei (2692.HK / 003021.SZ)		Inspire Robotics (unlisted) BrainCo (unlisted)
Dexterous Hand - Micro Ball Screw	Rongtai (603119.SS) Seenpin (unlisted)	Zhengyu (300953.SZ)		Inspire Robotics (unlisted) BrainCo (unlisted)
Dexterous Hand - Drive Module	Rongtai (603119.SS)			Inspire Robotics (unlisted) BrainCo (unlisted)
Dexterous Hand - Assembly	Everwin (300115.SZ) Seenpin (unlisted) Tuopu (601689.SS)			Inspire Robotics (unlisted) BrainCo (unlisted)
Rotary (Linear) Actuator Integration	Sanhua (2050.HK/002050.SZ) Tier-1 Supplier for Joints Tuopu (601689.SS)			
Head cover and glass	Lens (6613.HK/300433.SZ)			
ODM/Assembler	Tesla (TSLA.O)	UBTECH (9880.HK)	Bozhon (688097.SS)	

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Source: Citi Research, Company Reports

Figure 2. Chinese humanoid robot component makers which have set up plants outside of China

Company	Ticker	Humanoid robot component product	Plant location
Hengli Hydraulic	601100.SS	Planetary roller screw	Mexico
Leader Drive	688017.SS	Harmonic reducer	US
Rongtai	603119.SS	Micro ball screw	Thailand
Veichi	688698.SS	Micro motor	Thailand

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